COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010	**SECTION:**	CPE21S2
DATE PERFORMED:	07/15/2026	**DATE SUBMITTED:**	07/15/2026
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ACTIVITY NO. 2.1 Memory Diagram Exercise

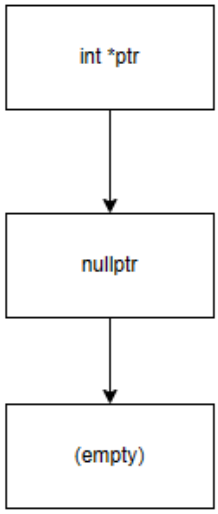
1. PROCEDURE OUTPUT

What procedures did you perform?

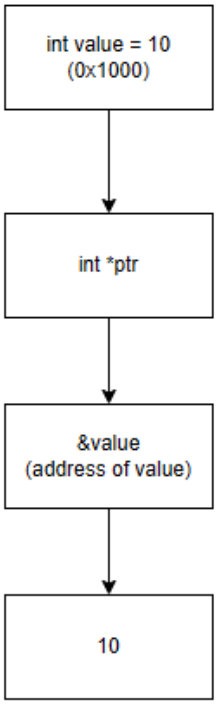
Draw the memory diagram for the following:

1. int *ptr = nullptr;
2. int value = 10;
   int *ptr= &value;
3. int *ptr = new int;
4. int *ptr = new int(10);

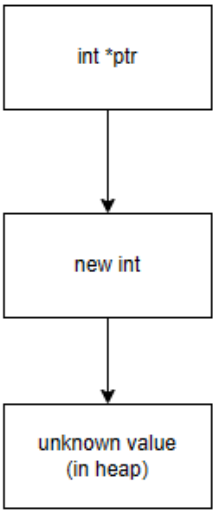
1.



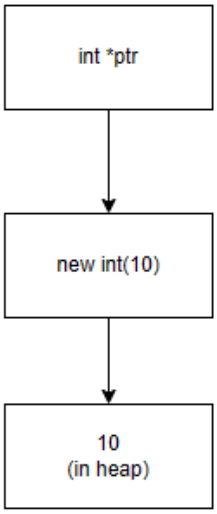
2.



3.



4.



5. Differentiate Constructor, Destructor, Copy Constructor, Copy Assignment Constructor.

The differences in the constructor and destructor come from how they behave. The constructor creates a new object by initializing the object and allocating it to memory, while the destructor deletes and flushes the allocated space used by the object created before. In other words, the constructor creates and declares the space to be used by a certain object, while the destructor does the other way around by flushing out the allocated space and freeing the memory space once allocated by the object. For the copy constructor and copy assignment constructor, the differences come from how they interact with objects. The copy constructor, by its name, creates a new copy of an existing object by allocating a new space for that copied object. On the other hand, the copy assignment constructor creates new data in an existing object by overwriting data in the assigned object using the data from another one.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **Google Gemini**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Summarization, explanation, and clarification of concepts in memory allocation based on the provided modules.**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **Used to provide insights and further information about the concepts in the class discussion.**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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